The completeness tables

The missing values as numbers: one line for the whole matrix, one row per sample, and one row per missing-value count.

AT A GLANCE

Where	Completeness → Tables
Data levels	Protein, phosphosite, phosphoprotein. Peptide and precursor are not offered here.
Needs	An uploaded dataset and its annotation.
Data source	The matrix as uploaded. There is no Filtered or Imputed choice on this page.
Shows	Three figures across the top, then three tables: per sample, the outliers among them, and per missing-value count.
Download	Each table separately, as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the page answers

Two questions that a figure answers only approximately. *How bad is it* — one percentage for the whole matrix, and one per sample, so that a run can be named rather than pointed at. And *which samples are past the line* — a threshold the user sets, applied to the per-sample percentage, with the samples that exceed it listed by name.

The three figures across the top are the whole-matrix summary:

- Overall Missing Values: the percentage of *cells* of the matrix that are missing, over every feature with at least one value and every sample at once.
- Outlier Threshold: the value currently set below, repeated so that the list beneath it can be read without scrolling back.
- Outlier Samples: how many samples exceed it.

The tables are recomputed when the level or the threshold changes, and the **Refresh** button re-runs them against the dataset as it now stands — after a collapse, a re-annotation or a new upload.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to describe: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run.	Protein	page
Toggle IDs	Labels the samples condition_n (id) rather than condition_n, where the id is the last number in the original sample name, or the whole name when it has no number.	on	page
Outlier Threshold (%)	The percentage of missing values above which a sample is listed as an outlier. Held between 0 and 100. The test is strict: a sample at exactly the threshold is not an outlier.	50	page

> NOTE
>
> This page draws no figure, so the plot appearance panel is not shown on it, and neither are the width, height, resolution and format controls that belong to a download of an image.

3 How the numbers are computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation and renamed as **Toggle IDs** requires. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing as well, unless the dataset was uploaded already on the $\log_2$ scale, where zero is an intensity of 1 and is a real measurement. Features with no value in any selected sample are removed first. Write m for the number of remaining features (rows), n for the number of samples (columns).
The line at the top. Every cell of the matrix, counted once:

$$\text{overall } \% \;=\; \frac{\#\{\text{ missing cells }\}}{m \times n} \times 100.$$

The sample table. Down each column, the count of missing values in that sample, over the number of features:

$$\%\text{ for sample } j \;=\; \frac{\#\{\, i \,:\, x_{ij} = \text{NA}\,\}}{m} \times 100.$$

The feature table. Across each row, the count of missing values for that feature; the features are then grouped by that count, and each group is given as a share of all features:

$$\%\text{ for count } k \;=\; \frac{\#\{\, i \,:\, m_i = k\,\}}{m} \times 100.$$

All three are rounded to two decimals. The percentages of the feature table sum to 100; those of the sample table do not, and are not meant to — each is a share of the same denominator, not a share of the missing values.

4 The sample table

Headed Sample-level Missing Values.

Columns in this table

Column	What it holds
Sample	The sample, labelled condition_n or condition_n (id), in the order of the annotation.
Counts	How many features are missing in that sample.
%_missing_values	That count as a percentage of all features, to two decimals.

5 The outlier table

Headed Outliers by Missing Value %. One column, Sample, and one row per sample whose %_missing_values is strictly greater than **Outlier Threshold (%)**. When no sample exceeds it the table is replaced by a line saying so, and the count at the top of the page reads zero.

Being past the threshold is not a verdict. It is the point at which a sample is worth looking at in the heatmap and the coverage plot: a sample that is missing 60 % of the features but is otherwise ordinary in the ones it has may still support some questions, while a lower overall percentage can still hide missingness concentrated in a scientifically important subset. Neither percentage alone decides exclusion.

6 The feature table

Headed Feature-level Missing Value Distribution.

Columns in this table

Column	What it holds
n_MissingProteinValues	A number of missing values, from 0 upwards. The column carries this name at every data level, not only for proteins.
Counts	How many features are missing in exactly that many samples.
%_missing_values	That count as a percentage of all features, to two decimals.

This table is the missing value plot, as numbers. Counts is the height of a bar and n_MissingProteinValues is its position on the x axis, with the percentage added. Both leave out the features that are missing in *every* sample, so no row here has n_MissingProteinValues equal to the number of samples.

The example worked

Three proteins over three samples, A1, A2 and B1:

THE MATRIX, AND THE THREE OUTPUTS

	A1	A2	B1
Protein 1	3.34	3.54	NA
Protein 2	3.34	3.54	2.67
Protein 3	3.34	NA	2.67

Two of the nine cells are missing, so the line at the top reads 22.22%.

Sample	Counts	%_missing_values
A1	0	0.0
A2	1	33.33
B1	1	33.33

n_MissingProteinValues	Counts	%_missing_values
0	1	33.33
1	2	66.67

At the default threshold of 50 the outlier table is empty: 33.33 is not greater than 50. Set the threshold to 30 and both A2 and B1 appear in it.

Edge cases

- A feature with no value in any selected sample is left out of this page — of the line at the top, of every sample's count and of the feature table — as it is left out of the missing value plot and the heatmap. A line under the cards says how many features were excluded. The notebook functions do the same in Python and R.
- A sample with no values at all reads 100.0 and is an outlier at any threshold below 100. At a threshold of exactly 100 nothing can ever be an outlier, because the test is strict.
- Both percentage columns carry the same name, %_missing_values, but they have different denominators — features per sample in one table, features per count in the other. The two downloads are separate files, so the names do not collide.

7 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
missing_protein_table(data, meta, id=False, data_scale="log2")
missing_sample_table(data, meta, id=False, data_scale="log2")
```

R

```
print(missing_protein_table(data, meta, id = FALSE, data_scale = "log2"))
print(missing_sample_table(data, meta, id = FALSE, data_scale = "log2"))
```

These two return frames rather than figures, which is why there is no fig = in front of them: what comes back is the feature table and the sample table, with the same three columns and the same names as the page, but unrounded — the two decimals are the page's, not the function's. The other two numbers on the page have their own functions: missing_value_outlier takes the frame that missing_sample_table returned and gives the names past the threshold, and all_miss_val gives the one percentage for the whole matrix.

8 Exporting

Each of the three tables has its own **Download CSV** button and downloads exactly what is on screen, with the same column names and the same rounding. The outlier table downloads its one Sample column; when it is empty there is no button, because there is nothing to write.

Behind this page

Function	What it produces
missing_sample_table	The per-sample table
missing_protein_table	The per-count table
missing_value_outlier	The samples past the threshold
all_miss_val	The one percentage for the whole matrix